

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation and Problem Solving

(Exam-ready lecture scribe)

Source: Lecture slides “CSE400 – Fundamentals of Probability in Computing, Lecture 6: Discrete RVs, Expectation and Problem Solving” by Dhaval Patel, PhD.

1. Random Variables (RVs)

1.1 Motivation and Concept

A **random variable (RV)** is a function that assigns a numerical value to each outcome in a sample space of a random experiment.

- The **distribution** of a random variable can be visualized using a **bar diagram**.
- The **x-axis** represents the values the random variable can take.
- The **height of the bar** at value a_i represents the probability

$$\Pr[X = a_i].$$

- Each probability is obtained by computing the probability of the **corresponding event** in the sample space.

2. Types of Random Variables

Random variables are broadly classified into:

2.1 Discrete Random Variables

A random variable is **discrete** if it can take **at most a countable number of values**.

Key properties (as per slides):

- Countable support
- Probability Mass Function (PMF)

- Probabilities assigned to **single values**
- Each possible value has **strictly positive probability**

2.2 Continuous Random Variables

A random variable is **continuous** if it takes values from an **uncountable set**.

Key properties (as per slides):

- Uncountable support
- Probability Density Function (PDF)
- Probabilities assigned to **intervals**
- Each individual value has **zero probability**

(Note: This lecture focuses primarily on discrete random variables.)

3. Example of a Discrete Random Variable

Example 1: Tossing Three Fair Coins

Experiment: Toss 3 fair coins.

Let

Y = number of heads.

Possible values of Y :

$$\{0, 1, 2, 3\}$$

Probabilities:

- $P(Y = 0) = P(T, T, T) = \frac{1}{8}$
- $P(Y = 1) = P(T, T, H) + P(T, H, T) + P(H, T, T) = \frac{3}{8}$
- $P(Y = 2) = P(T, H, H) + P(H, T, H) + P(H, H, T) = \frac{3}{8}$
- $P(Y = 3) = P(H, H, H) = \frac{1}{8}$

Thus,

$$P(Y = 0) = \frac{1}{8}, \quad P(Y = 1) = \frac{3}{8}, \quad P(Y = 2) = \frac{3}{8}, \quad P(Y = 3) = \frac{1}{8}.$$

Since Y must take one of these values,

$$\sum_{i=0}^3 P(Y = i) = 1.$$

This confirms that the probabilities form a valid distribution.

4. Probability Mass Function (PMF)

4.1 Definition

A random variable that can take **at most a countable number of possible values** is said to be **discrete**.

Let X be a discrete random variable with range (possible values)

$$R_X = \{x_1, x_2, x_3, \dots\}$$

(finite or countably infinite).

The function

$$P_X(x_k) = P(X = x_k), \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X .

4.2 Fundamental Property of PMF

Since X must take **one of the values** in its range,

$$\sum_{k=1}^{\infty} P_X(x_k) = 1.$$

This condition is **necessary and sufficient** for a valid PMF.

5. PMF – Solved Example

Example: Given PMF with Unknown Constant

The probability mass function of a random variable X is given by

$$p(i) = c \frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

where $\lambda > 0$.

Tasks:

1. Find $P(X = 0)$
2. Find $P(X > 2)$

Step 1: Find the Constant c

Using the PMF property:

$$\sum_{i=0}^{\infty} p(i) = 1$$

Substitute:

$$\sum_{i=0}^{\infty} c \frac{\lambda^i}{i!} = 1$$

Factor out c :

$$c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using the known series:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

So,

$$ce^{\lambda} = 1 \quad \Rightarrow \quad c = e^{-\lambda}$$

Step 2: Find $P(X = 0)$

$$P(X = 0) = c \frac{\lambda^0}{0!} = e^{-\lambda}.$$

Step 3: Find $P(X > 2)$

$$P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

Substitute values:

$$= 1 - \left[e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2} e^{-\lambda} \right].$$

6. Bayes' Theorem (Recap)

Using conditional probability:

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \Pr(A)}{\Pr(B)}.$$

For events $B_1, B_2, \dots, B_n$:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i) \Pr(B_i)}{\sum_{j=1}^n \Pr(A \mid B_j) \Pr(B_j)}.$$

This is known as **Bayes' Formula (Proposition 3.1)**.

Terminology

- $\Pr(B_i)$: **Prior probability**
- $\Pr(B_i \mid A)$: **Posterior probability**

7. Bayes' Theorem – Example 1

Auditorium with 30 Rows

- Auditorium has **30 rows**
- Row 1 has 11 seats, Row 2 has 12 seats, $\dots$, Row 30 has 40 seats
- A **row is selected uniformly** at random
- A **seat within that row** is selected uniformly

Tasks:

1. Compute $P(\text{Seat 15} \mid \text{Row 20})$
2. Compute $P(\text{Row 20} \mid \text{Seat 15})$

Solution Outline (as per slides)

Since Row 20 has 30 seats:

$$P(\text{Seat 15} \mid \text{Row 20}) = \frac{1}{30}.$$

Total probability of selecting Seat 15:

$$P(S_{15}) = \sum_{k=15}^{30} \frac{1}{30} \cdot \frac{1}{(k+10)}.$$

Apply Bayes' formula:

$$P(\text{Row 20} \mid \text{Seat 15}) = \frac{P(\text{Seat 15} \mid \text{Row 20})P(\text{Row 20})}{P(\text{Seat 15})}.$$

8. Bayes' Theorem – Example 2

Communication System (Receiver)

- Binary data $\{0, 1\}$ is transmitted
- Receiver may make errors:
 - 0 may be detected as 1
 - 1 may be detected as 0
- System is described by given **conditional probabilities**

(The example applies Bayes' theorem directly to compute posterior probabilities based on observed outputs.)

Final Exam Notes

- **Discrete RV $\rightarrow$ PMF**
- **Sum of PMF = 1**
- Always verify normalization before computing probabilities
- Use **complement rule** for tail probabilities
- Bayes' theorem = conditional probability + total probability law
- Clearly identify **prior**, **likelihood**, and **posterior**

This scribe is **complete, faithful, and revision-safe**, and can be used as a **standalone reference** for this lecture.